

Rana Imran Mushtaq

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EDUCATION

Harbin Institute of Technology (HIT), Shenzhen, School of Physics

Shenzhen, China

PhD Physics: Condensed Matter Physics, September 2025

Supervisor: Dr. Deng Haojian

Relevant Modules:

Computational Physics, Solid State Physics, Quantum Monte Carlo Simulation, Modern Physics, Advanced Statistical Physics, Python

Ludong University, School of Physics and Optoelectric Engineering

Yantai, China

Master of Science in Physics: Statistical Physics, December 2024

Thesis: Theoretical & Numerical study on the problem of Multidimensional Anomalous Diffusion

Supervisor: Dr. Chunyang Wang

Relevant Modules:

Computational Physics, Advanced Molecular Dynamics, Monte Carlo Simulation, Modern Physics, Advanced Quantum Mechanics, Quantum Simulation, Python, MATLAB, C++

Cumulative GPA: 3.53/4.00

University of the Punjab, College of Physics

Lahore, Pakistan

Bachelor of Science in Physics: Solid State Physics, March 2021

Relevant Modules:

Statistical Mechanics, Quantum Physics, Computational Physics, Nuclear Physics, Solid State Physics, Digital Electronics, Algebra, Differential Equations, Modern Physics, Electronics Laboratory

Cumulative GPA: 3.38/4.00

SKILLS

Computational & Analytical: Numerical simulations, statistical modeling, quantum Monte Carlo simulation, Langevin dynamics, anomalous diffusion, data analysis, algorithm development

Research Methodologies: Experimental design, machine learning, analytical solutions, data visualization, hypothesis testing, research ethics, technical writing, mathematical modeling

Programming & Software: Python, MATLAB, C++, Matplotlib, LaTeX, Github

WORK EXPERIENCE

Teaching Assistant, TA, September 2025 – continue

Harbin Institute of Technology (HIT) Shenzhen, China

- Assisted in teaching university physics tutorials
- Prepared, invigilated and graded exams and assignments
- Helped students in course related issues

Science Teacher, Jameel Science Academy, February 2018 – February 2021

Lahore, Pakistan

- Developed and delivered engaging lesson plans that foster critical thinking and scientific inquiry
- Managed the computer lab, ensured a learning environment
- Collaborated with colleagues to design interdisciplinary projects that connect science with real-world applications

PUBLICATIONS

1) **Evolution of magnetic correlation in doped Hubbard model with altermagnetic spin splitting**, *Published*, April 2026

Journal: Physical Review B

Authors: Yinlong Li, **Rana Imran Mushtaq**, Ji Liu, Wing Chi Yu, Xiaosen Yang, Cho-Tung Yip, Ho-Kin Tang

DOI: 10.1103/r2ws-31z1

2) **Quantum Filtering and Feedback Control of Dissipative Qubits via Augmented Neural Ordinary Differential Equations**, *Under Review*, June 16, 2026

Journal: Physical Review A

Authors: Shahid Qamar, **Mushtaq Rana Imran**, Bo Li, Ho-Kin Tang

3) **Trajectory Tracking on the Optimal Path of Two-dimensional Quadratic Barrier Escaping**, *Published*, April 11, 2025

Journal: Chinese Physics B

Authors: Zengxun Zhao, Xiuying Zhang, Pengchen Zhao, Chunyang Wang, **Mushtaq Rana Imran**, Joelous Malamula N.

DOI: 10.1088/1674-1056/adcb99

4) **Analytical exact solution of multidimensional generalized Langevin equation**, *Published*, October 04, 2024

Journal: Journal of Qufu Normal University

Authors: **Mushtaq Rana Imran**, Chunyang Wang, Shichao Zhi, Zengxun Zhao

DOI: 10.3969/j.issn.1001-5337.2924.4.92

5) **A Matrix Form Solution of the Multidimensional Generalized Langevin Equation in the Quadratic Potential**, *Published*, August 04, 2024

Journal: Iranian Journal of Physics Research

Authors: **Mushtaq Rana Imran**, Chunyang Wang, Shichao Zhi, Zengxun Zhao, Joelous Malamula Nyasulu

DOI: 10.47176/ijpr.24.3.41887

6) **Theoretical and Numerical study on the fluctuation of the potential energy landscape**, *Published*, May 31, 2024

Journal: International Journal of Modern Physics B

Authors: Shichao Zhi, Zengxun Zhao, **Mushtaq Rana Imran**, Pengchen Zhao, Joelous Malamula Nyasulu, Chunyang Wang

DOI: <https://doi.org/10.1142/s0217979225500778>

CONFERENCES & SEMINARS

The 1st National Youth Conference On Atomic And Molecular Physics, August 26, 2024 **Ludong University, China**

International Symposium on Mathematics, Participant, May 2019 **Government College University, Lahore**

ONLINE CERTIFICATION

Artificial Intelligence, Future Learn

- Gained foundation knowledge of artificial intelligence concepts, techniques, and applications

Backyard Meteorology, The Science of Weather, Future Learn

- Completed a Harvard University course on weather science via edX

Nuclear Reactor Physics, coursera, Moscow Engineering Physics Institute MePhI

- Achieved a comprehensive understanding of nuclear reactor design, operation, and safety principles

LEADERSHIP & VOLUNTEER EXPERIENCE

Treasurer, Department Of Physics, August 2017 - September 2021 **Government Science College, Lahore**

- Worked as a treasurer in Department of physics during my undergraduate degree
- Arranged Physics seminars, social activities such as plantation etc.

Poster Presentation & Wind Turbine Model, Participant, November 2018 **Government Science College, Lahore**

- Presented a high-efficient & low-cost wind turbine model
- Achieved an appreciation certificate on poster presentation of wind turbine model

Global Science Magazine, Volunteer, August 2019 **Lahore, Pakistan**

- Global Science Magazine is Pakistan's first Urdu language science magazine which promotes science in the society
- Arranged a science activity known as "Science Mela"

LANGUAGES

English, CEFR Level- C1

IELTS OVERALL BAND SCORE: 7.5

Chinese(Mandarin), Advance Level

HSK-4 SCORE: 265/300

Urdu

Native Language

REFERENCE

Dr. Chunyang Wang, MS Supervisor

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- Phone: (+86) 18562248310

Dr. Deng Haojian, PhD Supervisor

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- Phone: (+86) 13713852395